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SPEAKERS

Alex Gates, Speaker 1, Charles Levine

Charles Levine 00:00

So I'm going to Alex. I'm not going to be great at this, but I'm learning. I'm going to start and kind of tell you some background stuff that we know. Just bear with me, right? Because that kind of builds up. And then we'll go, Okay, so first of all, you know the core idea, we've got these competing forces. And then there's talk of the inverted U, right? And there's a lot of things that can contribute to this, but we're only going to start with the we're not going to formalize them all. We're going to start with the minimums, and then later on in the document, I'll say where we can go. Okay, cool. So like just talking about the first forces. You've got positive local forces, you know, development from stronger peers, higher training, competitive standards, Team context. You have stronger signals for performing in a high quality environment, more evaluated, retention, etc. And then use that negative local forces, right? And I'm going to talk about congestion, and we have congestion, crowding and congestion quality. That's not going to that's not part of tier one, but we're going to go there, and I'm going to explain it to you. There's a number of those. And then we also have individual level forces, your own performance, your ability, your role in the team, your effort persists, and stuff like that. And then the system level, right? We're going to talk a lot about global and local, right? And then there's just an example in part four, just kind of shows you examples of the local and global forces across all three domains. Now I put this note in your personal ability matters in all three of the domains, right? But we're, it's not going to come in, it's not going to come into the tier one thing yet, okay? Because it's going to be, it'll be a control that we put in there, right? So it's not ignored, but it's a background sorting force and so and we can use it as a control layer, or the first minimal force focuses on the local environment faced by the individual and global scarcity distinction. So this is talking about why we've got tier one and not tier one, believe it or not. This is very helpful for me. Part Six, it all boils down to in tier one, distinction is the thing that is the most, that is the scarcity quantity. I think it's distinction. And the word distinction has so much rolled up into it, right, right? So distinction is compounded of performance, opportunity to display that performance and recognition by evaluators or institutions, right? So we'll say, then we talk about an empirical outcome, that the capital y_1 , is you got drafted. Zero, you didn't, right? That's our thing. Now

we have that. So it's really three terms that I will talk about, the local term in a second, but this global scarcity is a capital lambda, right? And that global scarcity is, in this case, we're going to say it's the number of it's 60, the number of players that can be drafted by the NBA each season.

Alex Gates 02:45

Okay,

Charles Levine 02:45

okay, but it could also be how many people can be promoted, how many people can Dr, how many people can receive a lead evaluation, how much recognition the system can allocate, etc. And I'll talk more about extensibility in later models, right? Now we're going to talk about the local scarcity. The distinction is a capital, L, L, I, J, T, right? I is the individual, J is represents the team, and t represents the season, right? So what we've got is this local environment, including local opportunity, scarcity and delta, or the capital lambda, there is the global distinction. So the minimal formula is going to look like something like this, where we've got the probability of being drafted is a function of that local variable and the global distinction. Okay, now let's talk first. The first The first thing we're going to do is we're going to we're going to proxy that local environment with. We're going to pick something to proxy it with, and what we're going to proxy it with is Q. And Q is the congestion quality. In this case, Q stands for the average, the mean points per minute of your player team, minus you the leave one out, right? So that's going to be the first proxy for that logo, right? All right. I have some minimal assumptions down here, which you can look at, unless you want me to go over them

Alex Gates 04:13

well, so I want you to go and then I'll tell you my thoughts.

Charles Levine 04:18

Okay,

Alex Gates 04:18

yeah,

Charles Levine 04:20

then, because it's important. And I know this is something to you I just discussed you've got that internal tension inside the local inside the local term, right? Because really the local term, the net of it, is the benefits of that local term, minus the drawbacks or disadvantages of that term, right? And there's an example of some of the benefits, some of the drawbacks, and then later on, I can talk about what are the different specific data values that we could use to build those from the different domains. But clearly the changing is greater than the change in the benefits and drawbacks, right? So when you're to the left of this ideal queue, you're going to get more benefits are going to dominate to the right. That the drawbacks are going to and then the balance point is this q star, all right. So what I've done over here on the left is represent that the from the original plot, the five the for the original pipeline, the binnings that you're you're used to, showing those different biddings from lowest to highest, and then showing the probability of the outcome, right? And then on the right is that first fitted curve using, that's using linear probability model, right? And then we go on so then the first fitted model at the very bottom, that's

the real Biggie, right? You've got the probability of being drafted is going to be some alpha, which is the, you know, the baseline draft probability, plus a b1 and a b2 you see, I've taken that quality model and I've made it a quadratic,

Alex Gates 05:57

right? So what, what I don't necessarily like, is that this is built in the quadratic form from the beginning,

Charles Levine 06:05

okay,

Alex Gates 06:05

right? Like there feels like there needs to be a natural

Charles Levine 06:12

evolution

Speaker 1 06:12

to

Alex Gates 06:12

evolution to that without having that be because that's really a core assumption of the model, that's going to have a non linear shape, and we're going to fit that. And so that's just my first, first note thought, but a little bit more.

Charles Levine 06:29

Let me just ask you, if I'm new to this.

Alex Gates 06:34

Yep,

Charles Levine 06:34

we could still get a linear we could still get a linear model. The quadratic term would be zero, but we can't go from a linear model to a curved model.

Alex Gates 06:46

Yeah, all right,

Charles Levine 06:48

but clearly, I don't understand the way this really should evolve

Alex Gates 06:51

well. So it's not clear to me if the No, you don't need a quadratic to get the curve. Because really, what it could be is, is, you know, the non linearity can come from a different probability function. And so it can

be like a linear combination of two terms got a non linear suction. And so then it's about the trade off of the two linear terms

Speaker 1 07:14

got

Alex Gates 07:15

some added non linearity on outside of that, right? Yep. So it doesn't have to be,

Charles Levine 07:20

doesn't have to be one major variable with the quadratic,

Alex Gates 07:23

right, that type of thing. So let me step back and tell you a little bit more about what I see as our key points that I don't see here. I think the first thing that's important is to come up with what are the features of our system that are really important for the effect that we're seeing okay? And to think that through action is very helpful to think through the different things criteria we had when we were coming up with possible data resources for this, right? So what's, what's a bare, minimal model of promotion? Everybody has a inherent quality, some solid performance problems, yeah, and the probability of you getting promoted is proportional to your ability, right? That's, that's kind of like the that's

Charles Levine 08:26

a hopeful assumption,

Alex Gates 08:27

right? That's the bare, basic minimum type of thing. Yeah, right. Um, the next thing you could say is, well, that's the global you know thing, but what happens is, everybody's in local

Charles Levine 08:45

pools,

Alex Gates 08:46

pools, so without any peer interaction in your local pools, your probability of promotion is still proportional to your ability. No difference, right? And there shouldn't be any difference. But what happens when you do a instead of the the probability of promotion is directly proportional to your ability. It's now your ability rank, right? Then you start doing comparative stuff, moving from the raw problem, from raw ability to probability rank, introduces a comparative analysis without directly even comparing right, like it's just part, part of what ranking is, right?

Charles Levine 09:32

A local comparative

Alex Gates 09:33

analysis? Yes. And I think if everybody was still in a global pool, that change shouldn't have made a difference. So that's kind of where my mind is sitting, right. Like, why do we need the local pools? Why do we need the network effects to matter if everything is in a well stirred hetero, you know, system, global system, then actually, none of this should be, should be happening, right? Right, right? And that's that. Kind of that's my signpost of the minimal model has to capture that as one extreme case, whatever's going on here. That has to be a very clear thing that's in the model that we can explore, right? So to me, that's a core component of what's happening. And I don't see that kind of here, necessarily.

Charles Levine 10:23

If you look at the this one,

Alex Gates 10:25

yeah,

Charles Levine 10:26

we can set the local so the Q, those are the proxies for the L, the local effect.

Alex Gates 10:33

Yep.

Charles Levine 10:34

If we set that equal to zero, we get your probability we promote is somewhat proportional theta to your ability,

Alex Gates 10:42

good. So that that's the first, to me, that's the first thing we have to explore, right? Like somehow, that's the naive system that everybody wants to imagine.

Charles Levine 10:54

Yeah,

Alex Gates 10:54

right, yeah. In a true meritocracy, right? That would be what

Charles Levine 10:58

exactly

Alex Gates 10:59

is doing, yeah, right. And so that that clearly has to be a piece and a very identifiable piece of every model that we're building,

Charles Levine 11:06

okay?

Alex Gates 11:07

Is that that limit has to exist and be a component of it. And then the question in my mind is, what is the minimal ingredient you need to add to that to get the ranking thing that we have here. And I'm going to tell you what I think that, in my mind, that the two pieces are, one of them is the local comparison. But I almost think that even with the local comparison, if everything was still random, you wouldn't get it. What I think you need is the local comparison. And what you're terming here is congestion, which I kind of like, but to me, it's its correlation, that somehow you have some places where people are concentrating in their ability, that the the ability of your peers is more similar to your ability in general, and that disproportion the fact that we have concentration and then local ranking, are the two, only two elements that you need in order to produce this non linearity. Does that make sense? What I'm kind of saying, yeah.

Charles Levine 12:16

I'm just, I'm digesting it. It does,

Alex Gates 12:19

right? Yeah. So it's not even like congestion as like the measure of the things. It's actually, to me, the average of your peer group is a proxy for people want to have similar ability, want to come to these positions that are desirable, and the fact that you have some there is and that's the

Charles Levine 12:40

sort, that's the sorting that takes place the right? Yeah,

Alex Gates 12:43

right. So there's sorting, there's there's like a controlled sorting at the level of the groups. And so that's the element that we need to capture somehow. And I'm not sure what the right form of that is, right, but I think that's the that's the key, and that can be incorporated. You know, when I say I don't know the right form, there are different modeling techniques for incorporating that type of thing. But you could have, for example, a hierarchical model where you're first selecting the distribution of positions, and then you assign people to positions based on the prestige, perceived prestige, of those positions, or something like that, right? Like, that's the type of model that we need. Or there's a network based version of this. So

Charles Levine 13:31

I'm sorry, so you're saying that we need a model that kind of explains the pre sorting that takes place.

Alex Gates 13:37

That's because, to me, that's the key component. I guarantee you that if you come up with a model like this of local ranking, and there's not pre sorting of people, that the local ranking should still match the global ranking, I

Charles Levine 13:54

see what you're saying. If we just, if we just randomly sort people and put them into into player pools,

Alex Gates 13:58

yes,

Charles Levine 13:59

then even though we can still enumerate from zero to n , the player pools, the variance is going to it's not going to be a linear progression, and in the end, it's going to be a wash, I

Alex Gates 14:10

think so. And you can, you can do that,

Charles Levine 14:12

yeah,

Alex Gates 14:13

right, but I'm pretty sure that that's the case. Matter

Charles Levine 14:15

of fact, that should be one of the dotted lines,

Alex Gates 14:17

right,

Speaker 1 14:18

right?

Alex Gates 14:18

It might induce greater variance. So there might be a little bit more error because of that, right? But you should not get the non linear effect. You shouldn't get this dip at high so the key is these pools, one of the one of the

Charles Levine 14:38

traits of these pools, should be that, in general, the difference between the variance should be low of the talent of the talent in the pools, and the difference between your talent and the mean talent should be relatively small. Is that what you're saying you. Because that doesn't capture when we've got a superstar player that wants to go to Bible college,

Alex Gates 15:04

that's that's the variance of the pool. That's different.

Speaker 1 15:08

Yeah, I

Alex Gates 15:08

think that the quality of the pool is correlated, that the people who are inside of the pools tend to have similar levels of performance. So it's actually the opposite of what you're saying. I don't believe that the variance inside of the pools is super high. I think that people self sort

Charles Levine 15:28

No, no, I don't either. I said that the variance is relatively low inside the pool,

Alex Gates 15:32

yes.

Charles Levine 15:33

And the difference between the mean of that of the of the pool and the individual player is pretty small too.

Alex Gates 15:39

Yes,

Charles Levine 15:40

right?

Alex Gates 15:41

Yes, yes,

Charles Levine 15:41

that's what, that's what we're saying. But I think that that, and this is just we're talking and you're teaching, right? What about when I What about the big fish in a small pond? Because that happens, right?

Alex Gates 15:55

Well, so there's random sorting effects, right? Of course, this thing isn't set,

Charles Levine 16:00

okay,

Alex Gates 16:01

that there's random sorting effects. And the question is, how to incorporate that randomness in the right way so that you get the right types of distributions of these things that are realistic and and part of the question, so what you're

Charles Levine 16:14

saying is, we've got to introduce some sort of sorting noise,

Alex Gates 16:17

yeah,

Charles Levine 16:18

to get that randomness

Alex Gates 16:19

Right, right? So the, you know, like, here's a framework that I could imagine you have, right? You say there's some inherent quality of a local environment, so let's call that π_L , right? And this can be zero to one, and this is kind of like your inherent quality, right? And then you could say that the probability of an individual i being inside of some pool L is going to be proportional to the difference between the quality of i and the inherent the difference from that pools. Mean. Mean, yeah, right, something, well, no, so, like, this, isn't you pre selecting the mean? You're saying that that I'm going to first create a bunch of pools, and those pools are going to have some late quality feature associated with them, right? And then people are going to select which pool they want to go to, or some people are sorted into pools, right? I don't know the mechanism of the sorting. Yeah, right? So that that the probability of these things is proportional to this. And you can even, you know, you can just do a nice type of scenario, it's exponential, or something like that. Or it's cleavage, you know, Gaussian, around the difference of the pools. Or, you know, like, there's, yeah, there's ways of doing this type of thing, right?

Charles Levine 17:58

You're basically saying a BA model, where the you're going to be drawn to, right, the higher you are you're going to be drawn to, the more, the greater.

Alex Gates 18:06

And if you do this, if you look like the mean of this is that, so you're right, the this is technically defining the mean. If this is the probability distribution, that we're using these types of things, right? So there's, there's ways to kind of make it all meth fall out method,

Speaker 1 18:23

right?

Alex Gates 18:23

But this puts people into pools, and then you want to know the probability of a promotion which is your event. Why, given your own ability and something about the pool that you're in,

Charles Levine 18:40

right?

Alex Gates 18:40

Right? And my hypothesis is that if you do this, and for example, you make this unsort Like there's no here, this is just uniform between zero and one, or uniform equal probability of going into any of the pools, then this probability should be the same as if there was none the global right? But by putting people into local groups and having those local groups be correlated, we're now inducing enough effect where this becomes nonlinear in the pools, right to me, that's kind of the minimal elements. And then there's questions here, like, I think at the uniform level, this isn't enough. But what is like the most basic

probability distributions that induce this correlation that gets you enough to be able to do this, right? There's like, a whole series of things around that. Another thing that we've observed in real systems, but that may or may not be important, is that there's heterogeneous sizes of these groups, right? So, like, the size of group I is, you know, in some ways it could be power law. Depending on the domain, depending on the domain. Is that an important feature of the system, or is that just an extra complicating Yeah, I don't think that's necessary, because in basketball, the teams are about the same size. Yeah, we still get the effect. So I don't think that the group size heterogeneity is an important thing that we have to capture. It's a feature of the system, but it's not necessarily the thing. But this idea that there's congestion is what you're calling it, which, in my mind, is correlation. Is what the key correlation Tell me more. I just want to better understand what do you mean say correlation. What do you mean by

Speaker 1 20:34
that?

Charles Levine 20:35

When I say congestion, I'm talking about it's a broad term, and it's either congestion quality. Is that where the mean is a threat. Yeah, congestion crowding is the other one I came up with. Crowding is so like, please just forgive me for going with this. So let's say I own a soccer team, and I've got a I've got, you know, I'm on the left, I'm on the right side of that q star, and because I'm on the right side of that q star, I've got four players that are their talent levels off the charts, and I'm playing each of them \$10 million but I can't take advantage of that, because there's I've got, I would be better off replacing two of them with \$2 million players, because I can't take advantage of what they've got because of the because of the talent crowding.

Alex Gates 21:31

So, right? So I don't know the

Charles Levine 21:34

so why? The way I did that with this was to sum the points per minute of the players, not just take the mean

Alex Gates 21:42

Yeah,

Charles Levine 21:43

because that's a different dynamic. You could have a really high average if there's one more superstar on the team.

Alex Gates 21:48

So I would say to me, the correlation is that if I take the ability of Player I and the ability of player J, and I know that they're on the same team, I, then these two things somehow are kind of related to each other,

Speaker 1 22:06

right?

Alex Gates 22:06

That that there's a tendency for similarly performing people to be in the same group, even if these things are drawn uniformly.

Charles Levine 22:18

So that's a pretty What a pretty wild,

Alex Gates 22:20

right? And there's lots of different ways

Charles Levine 22:22

it's a wild homogeneity, but it's necessary to make this work.

Alex Gates 22:28

That's what I think. Yeah, right. And I don't know how it happens. And I think in different systems, it might happen differently,

Charles Levine 22:34

yeah,

Alex Gates 22:34

right. There might be, it's forced. So there's like, the normal class, the honors class, the super honors class, and some teacher told you which one you're in,

Charles Levine 22:45

yeah,

Alex Gates 22:46

right. In some systems, it could be immersion. So there's just this knowledge of what the prestige is of the different organizations and good people want to go to prestigious things. And so then they're aggregate like, yeah, the mechanism that induces this correlation, I don't know.

Charles Levine 23:05

I think that's just an assumption. We have to we otherwise we're just going to piddle around. We have to assume some stuff away.

Alex Gates 23:11

But I think what's key is that there is that correlation, that there is this differentiation between the pools.

Charles Levine 23:18

So, okay, great. So I'm just thinking about the army for lieutenants, there's zero correlation, yes, probably almost zero. They're pretty much randomly put places, but by the time they're captains and

choosing jobs as captains, they've got a track record, they've got a reputation, they've got some OERs. They know more about the army. Now they're seeking out the better jobs, and the better jobs are looking at their OERs and deciding whether or not to take them,

Alex Gates 23:44

yes,

Speaker 1 23:44

right?

Alex Gates 23:45

That's exactly.

Charles Levine 23:46

And in no one's going to go interview at Vanderbilt if they belong at Georgia State, right? Yep. And this is to them,

Alex Gates 23:55

right? So this whole thing, let's call it some q or what did you call it? Congestion? The SI

Charles Levine 24:01

sees it. See, yeah,

Alex Gates 24:02

okay, so it's called Si, right, like this is tunable, and it can take functional forms. Basically

Charles Levine 24:11

the cost

Alex Gates 24:11

there's, there's a bunch of different

Charles Levine 24:13

things, yeah, right.

Alex Gates 24:14

But somehow you should have a knob that you can turn here and say there's more or less of this. This is more this is a stronger effect. This is a weaker effect in the limit of this, not existing at all. The weakness effect, the non linearity, should disappear in the limit of this. Being super strong,

Charles Levine 24:34

we should get it.

Alex Gates 24:35

What do we I actually don't even, like, there should be a hypothesis. It's probably going up or down. Like,

Charles Levine 24:41

it's like we showed you when I selected only the teams that have draftees.

Alex Gates 24:45

Yes, it

Charles Levine 24:45

became monotonically decreasing,

Alex Gates 24:47

right, right? You get monotonic thing, but, yeah, but at the sweet spot in the middle where this effect is there, but it's not the dominant effect, you also have the interplay of,

Charles Levine 24:57

get that non linearity,

Alex Gates 24:58

right? Then you get the non. On here, and that's what I want. That's the sweet spot, right? And it shows that this, you know, the naive system is a limit. Yes, this is what people think. The thing that I'm interested in having is a inherently local interaction effect, like this locality correlation thing is inherently local. That can be interpreted as a network thing, by the way. So, like, there are network interpretations of what that's doing, right? I don't even know if we need that yet, right? Like, right, for me it might just be, it

Charles Levine 25:28

might still be something we paint later on

Alex Gates 25:30

top, right? But to me, it's an essential component, because I'm, I am pretty sure that that's the thing that's the magic sauce for this, that if all the groups were the same, that if we equally always distributed, that we wouldn't get the knowledge. So,

Charles Levine 25:46

so, okay, so what I'm hearing is, what we're really looking for is this, this probability is basically a function of some function of your your ability, your latent ability. Uh, and this theta, or this L this, this local thing, right? And if we want to do that in terms of linear programming or something, it's going to be something like this local thing, plus some theta, some proportional right? And if this goes to zero, then you just get the this, right, yep. And if this goes to a million, and then you get this. And what we're looking for is that something in between,

Alex Gates 26:33

right,

Charles Levine 26:34

right,

Alex Gates 26:34

yep, that's exactly.

Charles Levine 26:35

So this isn't all garbage,

Alex Gates 26:37

no, no, I think it's, it's kind of

Speaker 1 26:39

there,

Charles Levine 26:39

but it's saying I should start off with, hey, in a naive scent, in a naive environment where there's perfect evaluation and everything's unbiased, we've got the good Lord in His beneficence is looking into each player's art to see what they're worth. And we get a perfect correlated curve like that. And maybe it's but yeah,

Alex Gates 27:02

and I think it's this mixed with local comparisons, right? So again, the easiest way to make a local comparison thing is, is a ranking. So it's natural to want to just talk about rankings, which I think is completely fine, right? That's where,

Charles Levine 27:21

when you say ranking, though, okay with basketball, it's fine, because the pool sizes are the same.

Alex Gates 27:28

I'm saying ranking. So to that might be your scarcity thing in position here. So you could imagine that there's two different worlds, right? One world is, I don't care how many people I'm selecting, the probability that you're selected is proportional to your own ability,

Speaker 1 27:47

right?

Alex Gates 27:47

And if there's 100 people that are super good, right,

Speaker 1 27:50

all

Alex Gates 27:50

100 people get in, right? If there's 10 people that are super good, those 10 people get in, right, right? And so in that world, the probability of your promotion is just completely dependent on

Charles Levine 28:01

your own. Yeah. So that's No, no, no, lambda. Lambda is the is the global constraint,

Alex Gates 28:05

right? But you have this global constraint. You're only allowed to get in 20 people, 30, and I don't think it's the time. So one way you can have that happen is that there's a time selection thing, right? You're drawing each person in order, and there's maybe a random order or something like that. And so it matters where you are. The other way you do it is you say, Okay, I'm going to rank all 100 people, and I'm only going to take the top 10, right? That's your scarcity constraint. So I would go with the ranking. I think, to me, that's the natural

Charles Levine 28:39

the local ranking

Alex Gates 28:40

well. So the process is a ranking that can be a global ranking or a local ranking. But the process is not that I care about your ability. I care about the rank of your ability, right? And it can still be stochastic. It could be I rank everybody, and then there's still a roulette wheel draw so that, you know, maybe you get it, maybe you don't, right? It's like the actually the NHL draft, just the draft lottery just happened. And what they do is they do that all the teams get ranked from worst team in the league to

Charles Levine 29:18

best,

Alex Gates 29:19

best, and then the probability that you get the first pick in the draft, which is a very big deal, is proportional to your rank, but it's not

Charles Levine 29:30

stochastic.

Speaker 1 29:31

So

Alex Gates 29:32

like, the actually, the third worst team in the league got the bad first pick, is what happened, right? Yeah, because, but the worst team had the highest probability of getting it, but they didn't

Charles Levine 29:41

Right,

Alex Gates 29:42

right? And so the same thing can happen with this ability, right? You can imagine that I'm ranking all 100 people, and then I'm still flipping a coin.

Charles Levine 29:51

So do I so joining? I needed to introduce that stochastic nature into the to the ranking. Well,

Alex Gates 29:55

I think that that's, yeah, I think that the system is needs some. Some stochasticity, of course, yeah, to be fit, right? But to me, the key here is, why isn't the best person always selected? And some of that can be, it

Charles Levine 30:10

has to be explained by the by pools,

Alex Gates 30:15

no, so it doesn't. That's what I'm saying. The non linearity is the pool you

Charles Levine 30:20

said, Why isn't the best person always selected,

Alex Gates 30:22

right? And

Charles Levine 30:23

so I said, well, because it has that only explanation, that it's not a perfect linear thing is because of the pools.

Alex Gates 30:28

But even there it could be, it could be perfectly linear, and the best person isn't selected, right? You're getting the probability of selection

Charles Levine 30:35

right, right?

Alex Gates 30:36

Which is different,

Charles Levine 30:37

yeah, the logic pose it, yeah.

Alex Gates 30:40

So it's not that the first best person is always selected, but you have to have

Speaker 1 30:45

correct,

Charles Levine 30:45

right, right,

Alex Gates 30:46

right. And then you have to ask, what's the level of peer comparison? How was that element introduced? And to me, it's a very natural thing to introduce it just through ranking.

Charles Levine 30:57

And when you say a ranking, are you saying, let's do one to N of everybody in the and then maybe we multiply the probability times one over their ranking.

Alex Gates 31:06

Yeah. I mean, it doesn't really matter. Yeah, yeah, you'll see, I don't know. I don't Yeah, you could do something like that, right? But that's actually imposing already a functional form,

Charles Levine 31:16

yeah? Gotcha

Alex Gates 31:17

to me. What matters?

Speaker 1 31:18

What

Charles Levine 31:18

about, what about, what about their, their, difference positive or negative from the mean.

Alex Gates 31:26

Yeah, that

Charles Levine 31:27

still doesn't get after the ranking.

Alex Gates 31:30

So what I want to know is, look, there's 100 people in the in the thing, we have room for 10 people, right?

Charles Levine 31:44

You're saying globally, there's 100 people. There's room for 10,

Alex Gates 31:47

right? Yeah. So am I modeling that the probability that you're selected, which you put that I'm just putting sure like that, right? Is this proportional to your own ability, or is this proportional to the rank of your ability? That's the mechanism. This says, I know nothing about your peers,

Charles Levine 32:14

and that should be completely predictable,

Speaker 1 32:17

and

Alex Gates 32:17

that's what I think would happen. I don't think that the Local Group in this scenario would get you anything right again, right again. So this is another mechanism that you can actually explore. You can have a model where I'm not caring about the quality of my peers at all, even though peers correlate, and there's local blah, blah blah, if I don't have a comparison mechanism, there shouldn't be any peer effect. There shouldn't be any guarantee, right? In this scenario, there shouldn't be any non linearity. But in this scenario, now I'm comparing you against your peers,

Charles Levine 32:53

right?

Alex Gates 32:54

This is what should induce. This combined with the clustering, in local pools should induce Yes, because

Charles Levine 33:04

that represents the limits of cognitive human cognizance.

Alex Gates 33:08

We can

Charles Levine 33:08

only see so much. We can only concern ourselves with so much. And you know what I mean, nobody can see the big picture at all times, all they can see is what's going on around them. And even the people up here, they can't see the small picture, the teams. They can just see the things that rise to the top. Okay, I think, I mean, I think, I think I got you, I think we're thinking, I think we're thinking all the same things, all the I think we agree on everything that matters, just how this is just a way of looking at it. And so it's really just, I understand this. Now I have to go back and listen to the transcript, look at it and figure out, where do I go from here? I think I'm saying something like this, right? I'm building and saying the naive model is, is, is is, is, is, L is zero. L goes to zero. The local effect will say that whatever you want to call it. And then the next step is, how do I build this correct?

Alex Gates 34:11

Yes, yes, correct. So let's, I'm thinking about how to even operationalize this even more. Alright, so there's, there's two connections I want to make. The first is, what I really like about this formulation is this is very similar to the barbase Albert

Speaker 1 34:27

model.

Charles Levine 34:28

Did that that came to me right away, right,

Alex Gates 34:30

right? Yeah, we have two ingredients. Each ingredient is necessary combined. But if we don't have either one of them, we're not going to get it.

Charles Levine 34:39

You got to have a finite number of choices that you can attach to right?

Alex Gates 34:47

So I'm not sure actually about

Speaker 1 34:49

that.

Charles Levine 34:49

You don't know if that's important, but you're just saying that the if we're looking at the degree or the or the prestige of the teams, you're more likely to choose the more prestigious teams. But in the B. Gay model, there's no there's no fitness. The fitness is zero. It's a clean slate for the node choosing,

Alex Gates 35:06

yep, right. And I think that's the same thing here. Like, I think actually

Charles Levine 35:11

that would make sense for lieutenants, but

Alex Gates 35:13

I think that if the constraint that we can only select a few people doesn't even really matter, like you could actually put in here that your probability is proportional and stuff like that. And I don't care how many actually end up getting selected, but the fact that this induces a comparison and a distributional effect, no matter what, between the best and the worst, a differential effect between the best and the worst, like to me, this is what induces the differential, not the scarcity. I guess that's,

Charles Levine 35:46

yeah, no, you're right. Because if you look at the ER model, you're never going to have a max number of edges, right? It's going to take care of itself because of that probability.

Alex Gates 35:57

That's what I think happens.

Charles Levine 35:58

Yeah,

Alex Gates 35:58

that's, that's exactly

Charles Levine 36:00

that what

Alex Gates 36:00

you're saying, Yep, right. Okay. So what are the next steps on this? I actually think before we push further in terms of writing down an analytical model, why don't you code up an agent based model? Why don't you code this up as a simulation? And I'm pretty sure it's not going to be famous advisor, last words, that could be so hard, Charles, but, but here's the ingredients that I think that you need to

Charles Levine 36:25

have. Yeah,

Alex Gates 36:25

okay, let's, let's just use these nice numbers. Let's say that there are 100 people, right? And each one of them has a natural inherent ability that you select between zero and one.

Charles Levine 36:37

Okay,

Alex Gates 36:38

right? So 100 random numbers, that's their inherent ability, right? And then you want to know the probability of you being promotion. Promoted is just a nice normalized exponential of your own ability,

Charles Levine 36:57

okay,

Alex Gates 36:58

right? Something like this. This is, this is a classic physics thing to do, is that everything's these goops probabilities, and it works out. But you don't even need this. You could, you could just do the sum over all the things. I

Charles Levine 37:16

don't know what you just did. I saw that you did that exponential. You're saying basically e to the minus, whatever your zero to one is,

Alex Gates 37:24

yeah,

Charles Levine 37:25

over. What was the bottom one?

Alex Gates 37:27

You have to have a probability. So it's the sum over. Everybody's

Charles Levine 37:31

okay,

Alex Gates 37:32

right between zero and one,

Charles Levine 37:34

okay. And then why if you have the exponential of that one, it's just the exponential the top minus, the exponential

Alex Gates 37:43

this,

Charles Levine 37:43

yeah, the exponentials in both numerator and denominator. Why is that?

Alex Gates 37:46

Because this is the sum over everybody. If this is your own probability. So, so, yeah, let's we can talk about this for a sec. This is kind of important.

Charles Levine 38:01

Someone just told me about the Gibbs, something,

Alex Gates 38:04

yeah, the simplest model, yes. Forget about exponentials. The simplest model is, everybody has an inherent probability.

Speaker 1 38:11

I This

Alex Gates 38:12

is a number between zero and one. You have it for 100 people. You need to make it a probability so you divide by the sum over everybody's This is the normalizing term.

Charles Levine 38:23

Got it

Alex Gates 38:23

right. This becomes a probability.

Charles Levine 38:25

It's your so it's the α_i is just who you are. But yeah, α is that? That quantity zero to one

Alex Gates 38:30

exactly. So

Charles Levine 38:31

you take your α , right? Okay, got it

Alex Gates 38:32

right. So this is the probability of you being promoted given your ability is equal to your ability divided by the sum over everybody else's ability. So it's probability, right?

Speaker 1 38:44

Yep,

Alex Gates 38:44

this might work. This might be enough. But in statistical physics, they've learned that the difference between like So traditionally, it would be your energy levels or something like that, right? Like this is your energy level, that it's not this, it's e to this,

Charles Levine 39:08

divided by e to the sum,

Alex Gates 39:09

and then you need to normalize it, yeah, so it's the sum over it's whatever this is. It's the sum over that for everybody, is the normalizing term, right? Because you have to have numbers, but you have the whole vector.

Speaker 1 39:25

It's

Charles Levine 39:26

not e to the sum of the α s. It's the sum of e to the α Okay, yep, that's what I was confused by.

Alex Gates 39:32

Okay, anyway, so this is kind of a small detail. It might work enough with this, right?

Charles Levine 39:37

Yeah.

Alex Gates 39:38

Okay. Anyway, so now you have the probability that you're selected here. And you want to say that, like, let's just keep all the numbers nice, so 10 people get promoted. So ultimately, the probability that you're promoted is this. This is, you know, blah, blah, blah, and you can see that that, I bet you if you were to run this simulation. In 1000 times where you're selecting people, and you were to look, what's the probability of your alpha and the probability of you coming up, the whole system should be stupid linear. And you get exactly that, right?

Charles Levine 40:16

Where'd the 10 come from?

Alex Gates 40:17

10 is, I'm only selecting 10 people.

Charles Levine 40:19

Oh, so that's a global constraint.

Alex Gates 40:21

That's a global constraint,

Charles Levine 40:22

artificial, okay,

Alex Gates 40:23

yeah, and we can play with that and see, I don't

Charles Levine 40:26

know, yeah, no, I got

Alex Gates 40:27

either like, you could do another probability. If your probability is, like, greater than point seven or something like that. They're different thresholding techniques. This is a hard, this is a soft,

Charles Levine 40:41

yeah, whatever. Got it, got it, but it's an artificial constraint.

Alex Gates 40:45

It's an artificial constraint. Yeah, you're on the system 1000 times.

Charles Levine 40:49

Yeah,

Alex Gates 40:50

you should see something where the given your own ability. This happens,

Charles Levine 40:56

yep,

Alex Gates 40:56

right. Global thing. Okay, that's the first scenario, simplest,

Speaker 1 41:01

yep.

Alex Gates 41:01

Second scenario, you're now. So there's two components that we need. There's ranking and there's locality, right? So let's go locality. First

Charles Levine 41:13

locality meaning which pool

Alex Gates 41:15

right.

Charles Levine 41:15

Ranking is where you stand in that pool

Alex Gates 41:17

ranking is just that we're comparing you to peers, regardless of whether there's a pool or not.

Charles Levine 41:23

Okay,

Alex Gates 41:23

so that let's, let's walk through, let's see what's gonna happen. Okay, all right, so this is one, yep, two. Let's add pools. Okay, in pools, you now take your 100 people and you allocate them to some small number of pools, maybe 10, and they can all be equal sized. So really, what you're doing is you're saying I'm going to take 10 equal sized pools. Okay, actually, let's do that three. Let's do two. Is ranking, sorry, thinking on the fly. All

Speaker 1 42:14

right.

Charles Levine 42:15

Oh, so you're going to globally stack these people one to 100 to build the pools. Correct?

Alex Gates 42:20

No pools. Let's do rank everybody now. So it's the rank of alpha, and we'll call that gamma, just to make a global

Speaker 1 42:32

rank,

Alex Gates 42:32

your global rank,

Speaker 1 42:33

Yep,

Alex Gates 42:34

exactly. So the rank of alpha, I is that, and now I'm going to do the same thing. The probability of you being promoted, given your rank, is equal to your rank divided by the sum. Overall, it's just the probability.

Speaker 1 42:55

Yep,

Alex Gates 42:56

right. And again, you can do it the raw number. You can do it either the negative rank. Like, I don't know the right functional form for these things to kind of work out, but that's the first thing that you kind of have to figure out right to make this all work in this world. I think we should get the same thing that we got here. In other words, here's your rank, here's the probability of selection should just be,

Charles Levine 43:22

yeah, your rate just becomes a normal, normalized proxy for your alpha. Correct. Yeah,

Alex Gates 43:30

correct. Okay. Now the third thing is, we add pools. You group everybody into the 10. You do local ranking to equal size pools, you do local rank. And so that's the gamma of I in pool L, you do the same thing, probability of promotion given gamma i, L, and again, that's now. The one question is, how do you impose the normalization across pools?

Charles Levine 44:15

We haven't sorted

Alex Gates 44:17

what

Charles Levine 44:17

we haven't sorted

Alex Gates 44:18

haven't sorted yet.

Charles Levine 44:19

So you sorted, you establish pools, but without sorting you right?

Alex Gates 44:23

The pools are just you randomly assign each person to each

Charles Levine 44:26

okay,

Alex Gates 44:27

I don't know anything. There's no, there's no sorting to that process. Okay, I put you into a pool, right?

Charles Levine 44:33

Yeah. So this should just be the opposite of a bootstrap of that.

Speaker 1 44:37

That's

Alex Gates 44:38

what's gonna happen,

Charles Levine 44:39

right? It should be the exact same curve,

Alex Gates 44:41

yep,

Charles Levine 44:42

just like our reverse Bootstrap,

Alex Gates 44:44

yep.

Charles Levine 44:44

Okay,

Alex Gates 44:44

that's exactly right. But the trick here again, I don't know how you're normal. You're normalizing the problem, but you're right. The result is you should get the same thing again, right? And now more so we have. So this is your own ability, γ . I'm I , and this is what did I call

Speaker 1 45:03

What's

Charles Levine 45:03

your own rank? γ , I ,

Alex Gates 45:06

γ is your own rank, right, which is the rank of

Speaker 1 45:09

your ability.

Alex Gates 45:09

But now in this pooling scenario, you also have the PI of the pool, which is the average of the ability of people that went into that pool,

Charles Levine 45:24

okay,

Alex Gates 45:26

right for i and I , okay. So in the third scenario, we now have a new variable that we could look at, this thing in general, because everything was at randomly done should

Charles Levine 45:40

have very low variance, variance, and

Alex Gates 45:42

it should just be looking like everybody's point five, because everybody was pointed from a from high uniform distribution. But there's going to be some variability in it. So you could again plot here the PI of your L , and then the probability of promotion. And what we should see is, one, these things are really grouped over a small area, and two, that should just be full,

Charles Levine 46:09

yes, because the the mean of all those PI's are the PI is the mean the all those π should be pretty, pretty similar, because it's a random

Alex Gates 46:19

distribution. Yes. I mean the slight caveat is that, by knowing this, I know something about your ability. So maybe this is actually this.

Speaker 1 46:35

What do you

Charles Levine 46:37

how do you know anything about individual ability based on pies with no sorting.

Alex Gates 46:44

Great question. Maybe not. Could be that,

Charles Levine 46:48

because, I mean, we said we randomly sorted, so it should be flat,

Alex Gates 46:52

yep,

Charles Levine 46:53

right.

Alex Gates 46:54

So I actually, yeah, you're right. You're right. Should be that,

Charles Levine 46:58

yeah. Okay, so now,

Alex Gates 47:01

so here's, here's three simulations that you can run and build into to seeing the effect. And now we add the last thing

Charles Levine 47:11

so you're okay. So basically, just, just to make sure that water is wet, we're going to start with, boom, boom, boom. Okay, yeah,

Alex Gates 47:19

this is kind of how you want to be building

Charles Levine 47:22

in. Yeah, yeah.

Alex Gates 47:22

Okay.

Charles Levine 47:24

You are a physicist, man.

Alex Gates 47:27

So the last piece is the correlation. Now I don't know exactly how to do this, so my guess, let's just force the most extreme correlation that you can have. The most extreme correlation that you can have is that I'm going to put everybody into 10 pools. So let's erase a little bit and I'll write it down. And this might force I don't the reason I don't know like it might be too much is what I'm afraid of. But whatever it's the simplest I can think of at the moment,

Speaker 1 48:06

sure,

Alex Gates 48:07

I'll rank global

Speaker 1 48:11

on

Alex Gates 48:13

alpha, and

Charles Levine 48:14

then just put

Alex Gates 48:16

top 10 is pool one,

Charles Levine 48:18

yeah.

Alex Gates 48:20

Next is pool two

Charles Levine 48:22

Exactly.

Alex Gates 48:23

Force as much correlation as I can,

Speaker 1 48:25

yeah,

Alex Gates 48:25

into the pools. Now you do the whole thing,

Charles Levine 48:32

and you should see another linear relationship between Π ,

Alex Gates 48:37

yep

Charles Levine 48:38

and promotion,

Alex Gates 48:39

right,

Charles Levine 48:40

right, our selection for draft,

Alex Gates 48:42

yep,

Charles Levine 48:43

it'll be a little noisy, but not too noisy. It won't be noisy at all, not until we actually introduce some sort of noise to make this more

Alex Gates 48:51

now, yeah. So now if, if we do this actually, let's, let's, let's think about this all the way through. I gave you rankings. I've now eliminated the, really the influence of α . I actually think what's going to happen is, if I go back to this plot of α_i , and this, it should be almost,

Speaker 1 49:13

well,

Charles Levine 49:13

it depends on the distribution of the α .

Alex Gates 49:16

Yeah.

Charles Levine 49:16

What's that distribution of α_i ?

Alex Gates 49:19

Now, even if you're low α , is

Speaker 1 49:21

it

Charles Levine 49:21

a uniform zero to one? Is it

Alex Gates 49:23

that's it should still be uniform 01, but, but I think now if you're even if you're in a low alpha, I I've ranked you, and so it just matters. Are you the highest of the low

Speaker 1 49:35

you said we're

Charles Levine 49:36

talking about within our pool here. Okay, I thought we were talking about globally. What is the probability of your global selection based on your local rank? I mean, based on your local pool π is your local pool metric.

Alex Gates 49:51

So based on your local pool metric, it's still the same as here, but if you suck this all the way through with having done the global Alex. Location to pools, yeah, perfectly global. And you go back to your global alpha, I think we've eliminated this, this like preference. It's either gonna be something like this, or it's gonna be like a jigsaw, because it's about your rank within this group,

Charles Levine 50:21

but it's not you're talking about real world. We're talking about you're building this stuff in. If you think it's a saw tooth, that's because you built in.

Alex Gates 50:34

I'm in the model. I'm not in

Charles Levine 50:36

we got, we got one to 100

Alex Gates 50:37

Yeah,

Charles Levine 50:38

if we were to plot and if it's uniform distribution, we should have a 40 if we were to plot do a you know, number zero, right? Should be that well. Now we've arbitrarily done this 10 times, really, in effect by building up these perfectly sorted based on their and the rank. And since the rank is uniform, this should look really good, right? If it were uniform, if it were exponential, or something like that, we'd have some more craziness going on. But if it's uniform, we're gonna have something like that. So what I'm saying is, wherever my AI is isn't going to move, that's going to stay the same. So if I was here before this, now, I've got these values for probabilities. You see, because my local rank isn't going to matter, there's nothing that gages my local rank. We haven't introduced that yet.

Alex Gates 51:39

Okay? I've introduced it. That's the disconnect. Okay, I introduced it. You're right. And this is a great test case to see too.

Charles Levine 51:47

Yeah, you should just be the

Alex Gates 51:48

without having local rank too. So we use global rank to put you into a pool,

Charles Levine 51:53

yeah.

Alex Gates 51:53

And now you're saying, I don't have local rank. I just go back to my alpha. We should see no effect,

Speaker 1 51:59

right, right?

Alex Gates 52:00

But now put local rank back in. And

Charles Levine 52:02

if you're not looking at alpha, if you're looking at pi, then you're just going to have the same probability as the average person in your group.

Alex Gates 52:10

Yeah,

Charles Levine 52:11

right.

Alex Gates 52:12

Now put local rank back into it,

Charles Levine 52:14

okay,

Alex Gates 52:16

what happens if you put local rank back into

Charles Levine 52:19

well, because of the way that we did that, the local rank is completely transparent. Look, the local rank is just a subset. You see what I mean.

Alex Gates 52:29

But now your probability of promotion is based on your local rank, not your global.

Charles Levine 52:34

Ah, I see what you're saying. Okay,

Alex Gates 52:36

so that means that even

Speaker 1 52:38

if you're you're if you're if you're the top player in the bottom pool.

Charles Levine 52:42

So that's what the Marines do. The number of people, one to 100

Alex Gates 52:44

Yeah,

Charles Levine 52:45

the number number 100 gets his first choice of duty station. Number 90 gets his first choice. Number 80 gets his first choice. Number 91, is the last gets, you see what I'm saying?

Alex Gates 52:58

Yes. So they split it like that. That's what I think should have happened. And that might mean that the probability of promotion is something like that.

Charles Levine 53:05

So that's okay. Now we're saying that we've completely maximized the local effect, yes, so that's the naive maximization of local effect, right? Okay, and then yeah. And so what we should see as a step function, something, yeah,

Alex Gates 53:23

saw tooth function,

Charles Levine 53:25

yeah, okay, you're right, you're right, you're right,

Alex Gates 53:26

right,

Charles Levine 53:27

yes, yes, sawtooth. I like it. I got it.

Alex Gates 53:31

Okay. So then all these are the pieces. So then the last thing is, how do we get the non linear where does that come from? And I think the last piece is that it's a somehow, it's a competition between your own ability and the local rank of pools with correlation. Let's just call it local rank of correlated pools.

Charles Levine 54:15

Correlated means that there was some sorting

Alex Gates 54:17

right correlating or sort of I think if your probability of promotion is then proportional to these two factors somehow interplaying. So let's just even do the simplest version of that, then you're the each system is a probability of your own ability matters, plus the pool comparison matters. This is the final model that gets you

Charles Levine 54:57

normally

Alex Gates 54:58

right because. Well, if you think about it, this was this, and then this is this, and so they should well,

Charles Levine 55:06

then locally, the local rank is key, because the local rank, again, washes out a shitload of all this stuff, right? The local rank accounts for the pressure. The local rank accounts for, no, it doesn't account for the pressure. It doesn't, but it's a well, yeah, it does, because if you're at the bottom of the top, but something bothers me about this, and it should, the sorting should take care of that. And the good news is, in the real world, there's going to be a stochastic sorting,

Alex Gates 55:40

right? And so it could actually be that this hard sorting, this global hard sorting, is actually the limit case of something more interesting, and it might be too extreme. That's that in my pit in my stomach, is that that's the the missing component. So how do you put people into groups in a realistic way that's going to get this effect. And does it matter how we put people into groups? Let

Charles Levine 56:06

me see if I heard you correctly. What you're saying is this naive total like I told Peter, some people have all their students. They had to put them in groups. And they just go one to N and they chop, chop, chop, chop. Those are the elitists that do that. I usually take the top guy and the bottom guy and the and like this, right? But even that has but you're saying that if there's no way that we could justify even using as a model this maximum one to n, right? Because I just can't see where that's going to happen in nature. Is that what you're saying,

Alex Gates 56:41

Charles, you're brilliant.

Charles Levine 56:42

I know that. I just want you to know it. Yes,

Alex Gates 56:46

yes. Maybe the whole thing is about that.

Charles Levine 56:54

The whole thing is so, like, how

Alex Gates 56:57

did we put the people into the group, exactly,

Charles Levine 56:59

and the army, and it's like the Duke University is not going to hire 10 lazlos, because they'll be over budget within 10 minutes and and they're not going to hire 10. What does that guy, the Russian guy, that

Alex Gates 57:20

Dimitri,

Charles Levine 57:21

Dmitry, you're not gonna hire 10 of those guys either. They're gonna there's gonna be inherent diversity, right?

Alex Gates 57:28

Yes.

Charles Levine 57:28

And again, being a good player is not about points per minute, it's about a whole shitload of stuff. And some people are good at this, and some people are good at that. That all goes into it, and that's why we have this stochastic effect in the sorting. And it's not uniform, not a perfectly ordered,

Alex Gates 57:51

right?

Charles Levine 57:52

I can't like so University is going to do that. Players are going to do that on the coaches are going to do that on the team. The army is not going to put all of, if they can help it, they're not going to put all of the same kind of people into the organization, right?

Alex Gates 58:05

Yep.

Charles Levine 58:07

So why is that brilliant? Why is that

Alex Gates 58:10

because that's not the only way that people get sorted. So the systems we've looked at, we've selected it specifically because underlying it, there's this idea of prestige sorting,

Speaker 1 58:23

right?

Alex Gates 58:24

That was our whole entrance into this problem. Was this idea of prestige sorting. What happens if you have different sorting mechanisms? Can you get different shapes of these functions happening if you change so the prestige sorting is like one version, but what you just said, or you just articulated two other versions of this, right? There's the hard, prestigious, like a really hard top affiliation sorting right, and then there's top and bottom

Charles Levine 58:58

right.

Alex Gates 58:59

And in a version of this, we also had random, so there's already three, three versions of sorting, yeah, that you can use to form your groups in this scenario, that everybody has some inherent ability, and the the way that you formed groups is kind of dependent on those abilities, top, bottom,

Charles Levine 59:24

yeah. What we're really saying? Yeah, I think we can. I think it's almost a given that you're going to get the U provided they're sorted assortatively. There's assortative mixing.

Alex Gates 59:38

That's exactly what the story lines. Correct. Correct,

Speaker 1 59:44

right?

Alex Gates 59:44

That's what? Yes. So that's the underlying assumption.

Charles Levine 59:47

I hate when this happens before the dissertation is complete. What do you mean? Because I felt like I had a clear path of where I was going, and now it's 10 times more interesting. Of 10 times more profound and 10 times more challenging.

Alex Gates 1:00:04

I don't think we have to take it in all these directions yet.

Charles Levine 1:00:08

Yeah,

Alex Gates 1:00:08

right. I think now is the right. It's still playing with the model. It's still pushing a few of these assumptions. It could actually be that no matter like, we only have the assortative I like the assortative sorting. I think that's a great term for the for the effect. So you have assorted of sorting, and you have other ways of sorting, line sorting, the sorting, of sorting, I'm almost afraid some of these other ways might end up being equivalent to each other and not get you any of the effects that we want now that we're thinking about it, but, but there's a big question there, if you had disassorted A sorting, what would it look like?

Charles Levine 1:00:43

That's the zero and ones, yeah,

Alex Gates 1:00:45

if you had meat, like, I don't know, like, what are the other types of a sorting? So

Charles Levine 1:00:52

I was just thinking, What? What if everybody, they've got, what if everybody, when they made captain, the army sent them to the base closest to their hometown? I mean, you know, stuff like that, right, right? Some, some method. But,

Alex Gates 1:01:06

well, maybe actually, that's it. There's assortative disorder and random,

Charles Levine 1:01:10

right? And disassociative, is it really just sort of, because you can get close to the middle, it becomes more sort

Speaker 1 1:01:16

of

Alex Gates 1:01:16

assortative, yeah, disassortative and random might be the same.

Charles Levine 1:01:22

Um, I think it's going to be like something like this, where it's going to be like no effect in the middle and more effect in the tails. So cool, when you're putting this guy and this guy together, your your this assortitivity is super high, yeah, but it's going to linearly decrease as you get to hear, yeah, right, great.

Alex Gates 1:01:44

Great.

Charles Levine 1:01:45

Why is that great?

Alex Gates 1:01:46

Because it's three different effects,

Charles Levine 1:01:49

assorted, yeah, because that's it, that this assertivity is, it's not what we want. It's just different, yeah, yeah,

Alex Gates 1:01:56

no, this is really great,

Speaker 1 1:01:58

is it?

Alex Gates 1:01:59

Yes? So this is, this is the type of model that I think we could really run.

Charles Levine 1:02:04

Alex, 25 things like this come to me every day, but I don't have the ability yet to sort them or know which is relevant, which isn't, and which is not just some radical it's changing. But, yeah,

Alex Gates 1:02:18

you'll develop your intuition. So this is great. So I do think that the right thing to do now is try running these simulations with these basic assumptions that we just laid out, and verify that the functional forms that we're getting is right.

Charles Levine 1:02:35

Okay, to run these simulations. Then let me just make sure I'm tracking I'm still going to rely on, I'm going to, I'm going to give everybody uniform from zero to one for alpha, right,

Alex Gates 1:02:50

yep.

Charles Levine 1:02:50

And then I'm going to, I'm going to need to rank them zero to n for ranking.

Alex Gates 1:02:56

Yeah,

Charles Levine 1:02:56

right. And then I'm going to, I'm still going to say that the probability of, okay, I need to think more about this. Let me, let me go short my pencil and come back to you with questions. But basically, you say, run a simulation. You're basically saying stochastically, introduce a little bit of of stochasticness or randomness. But basically, if it was, there was zero randomness, it would either be this perfect, this assortitivity, perfect assortitivity, or random.

Alex Gates 1:03:34

Yeah,

Charles Levine 1:03:34

right. And then you're going to say, okay, that's how I've grouped them. Now that's the, this is the part where I'm not sure where to go next, then I need to run simulations to test if they get promoted, right?

Alex Gates 1:03:46

Yeah.

Charles Levine 1:03:47

And how do I do and that whether they get promoted or not is based on

Alex Gates 1:03:51

there. So again, it's, it's these functions, y, probability of Y, yeah, right. That's probability of promotion, right? So in the first one, you have the probability of promotion purely based on your own,

Charles Levine 1:04:05

yeah,

Alex Gates 1:04:07

right. And then we're starting to add these other things, and you're looking at the relationship between your rank, your your alpha, your inherent ability, or the rank of your ability, and the oh, okay, got

Charles Levine 1:04:21

so this is still, still holes, right? But then we're putting in, well, I'll start with additive that we're putting in this other term here.

Alex Gates 1:04:29

Yeah. So, so I think before you do the final model, do these pieces first that we just laid out, and to

Speaker 1 1:04:36

go

Charles Levine 1:04:36

sit down and make sure I'm trying. Because I think that when you said, do the simulation. I haven't done a simulation in eight years.

Alex Gates 1:04:43

Okay,

Charles Levine 1:04:44

so I need to, like, figure out exactly what you mean by that. Yeah. I mean, I know what you mean by it, but connecting. Just give me some time to

Alex Gates 1:04:53

cool,

Charles Levine 1:04:54

great.

Alex Gates 1:04:55

Try to run. Run the thing by the box too. And. You can kind of by Claude, or whatever you want to use, and the transcript of our conversation by Claude, have it pull out and kind of map into, what are the simulation components that have to happen here, and the layering of those simulation on top of each other? Yeah, right. And again, it's building up to this effect and the graphs that we need to characterize each simulation should be a probability of your promotion based on your ability, a probability of your promotion based on the rank of your ability, and then a probability of promotion based on something about your pool,

Charles Levine 1:05:38

much clearer,

Alex Gates 1:05:40

right? And you have those three graphs. You can't do the pool for the first two or three until you pool them. Once you pool them, you have all three. And then you add in these extra assumptions, and you run 1000 times of these mock populations that you're building, and you look at what ultimately the probability of promotion and so

Charles Levine 1:06:05

what we're really saying is it's the sorting that creates the curve

Alex Gates 1:06:12

correct,

Charles Levine 1:06:13

and we've been able to show empirically that the curve is au So now what we're really doing is saying something about how these people were sorted,

Alex Gates 1:06:20

yes, that would be our insight. That's

Charles Levine 1:06:23

fucking cool.

Alex Gates 1:06:24

That would be our insight. And if this all works out in a simple way, then there should be a way to take this model and treat it analytically. But I don't want you to go down that rabbit hole yet, because I don't know if it's going to work, and it's much easier to do the simulations than go analytically. Right?

Charles Levine 1:06:40

Agreed, yeah, if

Alex Gates 1:06:41

this works, then, given some assumptions about the I don't like, I don't know if the uniform probability distribution, then, is the one we want to use to treat it analytically, or what, like, you'd work that. But there would be a way to treat this analytically, and then you're going to get a analytical thing that you can fit to the actual data that we have, and then we'd know what is the relative versions of these things and the trade off between them based on The data that we're actually seeing.

Charles Levine 1:07:17

Nice,

Speaker 1 1:07:18

yes,

Charles Levine 1:07:19

yes, yes. It's one of those things where, like, now you need to get

Alex Gates 1:07:25

so right. We don't know if this is going to work

Charles Levine 1:07:29

right?

Alex Gates 1:07:30

The big gaps in my mind that easily could point to why this isn't going to work is one, this idea. Where did it go? The

Charles Levine 1:07:41

uniform distribution?

Alex Gates 1:07:42

Yeah, it's

Charles Levine 1:07:43

right

Speaker 1 1:07:43

there.

Alex Gates 1:07:43

So this was a pool size, but we also did a uniform distribution of your ability.

Charles Levine 1:07:48

Yeah, it could be, it could be, it could be bimodal, with humps in the either end. It could be normal with the hump in the middle,

Alex Gates 1:07:54

right? Yeah. So what's the shape of this? Right? That that we need, and

Charles Levine 1:07:59

if we want to know that, we can find it by getting the average GP you can get GPAs right, the officers. We can get, you know, number of citations or papers before they we can find some metric here, yeah,

Alex Gates 1:08:13

and it might not be dependent on it, because if, across all these different systems, you know, publications is heavy tail, the number of your own top block ranks is, has a weird distribution, and then points per minute is, can't be heavy tail, like, that's capped, right, right? Oh, I don't know. Like, that's probably something like a, yeah. Anyway, so so it might not matter. So there's the hope. But that's the first thing, yeah. The second thing is this probability of just being, like the dumb probability, where you just add up and divide by the sum, versus something more complex, like a Gibbs probability, versus something even more complex than this, that I don't know what's the right weight, right, right? But that's

Charles Levine 1:09:01

what GPT is for, right?

Alex Gates 1:09:03

This is the second choke point that I see. The third choke point is this thing, this, this idea of the sort of sorting, and how do we have a knob that controls the amount of assortativity, or disassortativity in our sorting.

Charles Levine 1:09:23

Oh, I think I could mathematically pull that out of my ass. Cool,

Alex Gates 1:09:26

where that that piece of it, like, if we just,

Charles Levine 1:09:32

I mean, you could. You could literally do it the the ER or the BA method. You just keep running it until everybody it's like, who's next, whatever the thing you know, who's next,

Alex Gates 1:09:44

great. And

Charles Levine 1:09:45

you just keep going until you're out of people. I've done all 10 Network

Alex Gates 1:09:48

form growth formulation, right? Is this problem? So you can kind of treat it that way, and that's where this is a network PhD, like, there's network based models that would give you this. With a different level of assortative.

Charles Levine 1:10:01

Please remember what you just said, so that you can remind me of it later. Yeah, but, I mean, I think that's easy. You say, you just give a problem, you know, this is your rank and the number, and so whatever your you got it, and so your probability, it could be the number two guy that's going to get the next slot right, but it's most probably going to be you. And then you whatever you have left. You run it again. You keep doing that until you get all 10 stacked. That's as simple.

Alex Gates 1:10:27

And so then the only thing that all of this framework doesn't capture, which we thought was kind of important in all of the work we've done so far, is that we also have a competing attrition. And none of this has attrition built into it. So the last stumbling block could be, none of this actually matters, and the secret ingredient was actually the attrition, and it's the competition between somehow attrition is related to your performance as well that actually gets you the non linear so that's the other modeling framework that this could go down. What I don't think is that there's

Charles Levine 1:11:05

a linear relationship, or a complete, proportional or direct relationship between attrition and rank,

Alex Gates 1:11:13

yep.

Charles Levine 1:11:13

So that takes a lot of that out of it. I think

Alex Gates 1:11:15

good,

Charles Levine 1:11:16

because the because attrition just changes your rank automatically, no matter what,

Alex Gates 1:11:20

right,

Charles Levine 1:11:21

right,

Alex Gates 1:11:21

yep, and the fact that we've seen this across different domains where there shouldn't be the same kind of attrition relationships is good like, would you have imagined that the best basketball players are also not likely to Play basketball anymore,

Charles Levine 1:11:40

right,

Alex Gates 1:11:42

right?

Charles Levine 1:11:42

But I mean, if you ask somebody on the street and you said, Hey, do you think the entire senior class at Duke University is going to get drafted? And they'll say, No, but that's the naive expectation,

Alex Gates 1:11:54

right? But do you think that the best person at Duke might have some incentive not to keep playing basketball? That's the attrition effect that I'm I'm worried about,

Charles Levine 1:12:03

yeah, oh, right, you're Yeah, because he's going to be a great doctor,

Alex Gates 1:12:06

right?

Speaker 1 1:12:07

Yeah,

Alex Gates 1:12:07

maybe, or

Charles Levine 1:12:08

whatever, but he's

Alex Gates 1:12:08

going to go make his money in the NBA if he's really the best player at Duke, right? Like, I don't know that's, that's the attrition effect. So this is the other reason that this whole modeling framework might not work,

Charles Levine 1:12:19

because it did, because they aren't closed domains,

Alex Gates 1:12:21

right?

Speaker 1 1:12:21

I mean,

Charles Levine 1:12:25

that's definitely the case in the army, some of the best guys leave to go make more money.

Alex Gates 1:12:29

So that's the question. Is that the driving effect, or is it the stack drinking local pool thing?

Charles Levine 1:12:36

Oh, what you're saying is elementary Yeah, I see what you're saying. I I think leaving that you're saying, is it the stack ranking effect, or is the attrition effect doing it? I'm saying that you cannot disambiguate the two. They're the same phenomenon.

Speaker 1 1:13:00

You

Charles Levine 1:13:01

see what I'm saying? It's, it does it's a wash. It doesn't matter, because that's always gonna, I feel like that's always going to be there, and they're always going to be this two sides of the same coin.

Alex Gates 1:13:16

Well, I'm saying that in basketball, I don't think the attrition effect is the same as the one in the army.

Charles Levine 1:13:23

Oh, right.

Alex Gates 1:13:23

And I also think kind of that academia might be that way too, that the very best person in a department is probably likely to stay in academia,

Charles Levine 1:13:37

but the very best person in a department, if it's clear that, for political reasons,

Alex Gates 1:13:42

might move, she

Charles Levine 1:13:43

might move to another department or another institution,

Alex Gates 1:13:46

right?

Charles Levine 1:13:47

You see,

Alex Gates 1:13:47

is it the same thing in basketball?

Charles Levine 1:13:49

Yes, I'm not getting enough playing time.

Alex Gates 1:13:51

So then actually, all those systems could be the other driving Yeah. So, so I'm just explaining to you what the what my

Charles Levine 1:13:57

classmates that went to, that came to the Citadel a year after me from high school,

Alex Gates 1:14:02

yeah? Because

Charles Levine 1:14:02

the where they went first, they weren't getting enough playing time, and the citizens team sucked,

Alex Gates 1:14:06

right?

Charles Levine 1:14:06

Yeah.

Alex Gates 1:14:07

So just to summarize this, then at very, very high level, this is, I think one modeling direction. I think this is the more promising, more interesting one from the network science point of view,

Speaker 1 1:14:21

yeah,

Alex Gates 1:14:22

so I hope that this is what works, right, and is enough to create the effect,

Charles Levine 1:14:26

yeah.

Alex Gates 1:14:26

The other modeling framework that we might need is something about attrition, and it's just the trade off with attrition,

Charles Levine 1:14:36

something to explain that dynamic,

Alex Gates 1:14:38

correct,

Charles Levine 1:14:38

that can be plugged in later.

Alex Gates 1:14:40

Yes,

Charles Levine 1:14:41

right? Or can further decompose it?

Alex Gates 1:14:43

No, no. Like, the whole non linearity effect is driven by some attrition correlation underlying Yeah,

Charles Levine 1:14:49

oh, wow, yeah,

Alex Gates 1:14:51

right, the competition, like, the whole modeling framework, could just be that it's this competition between your own ability and why you think you're going to be promoted. Yeah, some attrition. That's. Based on your ability that competes with it, and that means that in the top pools, you know, like, that's that's the other framework that that could explain the effect. And that one, to me, is slightly less interesting.

Charles Levine 1:15:15

What's the attrition like in the top pool versus the attrition in the

Alex Gates 1:15:19

correct

Charles Levine 1:15:20

empirically, I showed you it's much higher,

Alex Gates 1:15:22

yes,

Charles Levine 1:15:23

much higher the top pool than the bottom pool.

Alex Gates 1:15:25

So is that enough to just get the probability of your promotion being less up there

Charles Levine 1:15:31

too? Oh, fuck, right, because you can't be promoted if you quit,

Alex Gates 1:15:36

yeah, but the survival analysis should have taken that into account,

Charles Levine 1:15:41

but, but then again, we're

Alex Gates 1:15:43

the survival analysis should have taken that into account. That was the whole point of the

Speaker 1 1:15:47

survival, yeah,

Alex Gates 1:15:47

but maybe it didn't take it into

Speaker 1 1:15:49

account.

Charles Levine 1:15:50

Well, with the competing risk, yeah, it took it with competing risks. It took it into account by showing us both,

Alex Gates 1:15:57

yeah,

Charles Levine 1:15:57

the Kaplan Meier curve did not take it into account. Great. So But Alex, step back, what are we asking ourselves? We're asking ourselves, what do you need to do to win games? If you're How do you need to sort to win games if you're a coach or if you're an individual? What do you need to do to go to the NBA? We're not asking about those people that decided not to go to the NBA. So really, the question we're asking is, how do you get to the NBA? We're not explaining why people quit. You see what I'm saying? We're saying, how do you get to the NBA?

Speaker 1 1:16:33

Yeah.

Charles Levine 1:16:34

So if people if there's attrition going on, that's great. But you didn't ask, How can I be happy? You asked, How can I get to the NBA? Those people are always going to be there that decide the NBA is not as important as this.

Alex Gates 1:16:49

Yes,

Charles Levine 1:16:50

right,

Alex Gates 1:16:50

right,

Charles Levine 1:16:50

but that doesn't concern you. If your question is, how do I get to the NBA?

Alex Gates 1:16:55

Yes, true. So hopefully that's not

Charles Levine 1:17:00

so if we're selling this to General Dynamics, we don't have to worry about what you just said. But if we're selling this to science, which I think this paper is,

Alex Gates 1:17:07

well, that's what this

Speaker 1 1:17:08

is,

Alex Gates 1:17:09

yeah. But optimistically, I think that this idea of assortative sorting into local pools and probably a promotion is then comparative rank in your sortatively, sort of promotional pools is what drives this effect. Is a very nice, clean narrative. So from a God, I hope this works standpoint, and I'm very excited

Speaker 1 1:17:37

about

Alex Gates 1:17:37

the potential.

Charles Levine 1:17:40

Yeah, I am too. I'm really excited. Like, I'm on a whole nother plane. This is when all these things start to connect. You know?

Alex Gates 1:17:48

Yep,

Charles Levine 1:17:49

like, I know all this math shit and like, oh, it's, it's all useful,

Alex Gates 1:17:53

yeah? So, so here's what I'm gonna say. I think, realistically speaking, doing the first two or three simulations, you should be able to do that in the next two or three days, maybe over the weekend. You might not get all the way through all of them, but that's okay. Let's talk on Monday or Tuesday, either one

Charles Levine 1:18:14

you're saying, just simulate probability based on global rank and probability place based on no you're saying probability based on latent ability and probability based on the global rank of your latent ability.

Speaker 1 1:18:29

Yeah. And

Charles Levine 1:18:30

then we'll start to begin to build that,

Alex Gates 1:18:31

right,

Speaker 1 1:18:32

yeah. Okay.

Alex Gates 1:18:33

How long does it take you to

Charles Levine 1:18:34

do that? I think you're, you're right on the money, yeah, yeah,

Alex Gates 1:18:36

right. And then we'll make sure that this is doing what it's supposed to be doing,

Speaker 1 1:18:40

yep. So touch

Alex Gates 1:18:41

base. So don't think like you have to get all the way through before we talk again.

Charles Levine 1:18:44

Yeah. Please don't read my face as happy or not happy. I'm static right now. Cool. Does

Alex Gates 1:18:51

that make sense? Yes. Like, let's, let's have a check in point. After some simulations going and you started to tweak it,

Charles Levine 1:18:56

yeah, and

Alex Gates 1:18:57

then you'll see how easy it is. Are you

Charles Levine 1:18:58

gonna remember all this?

Alex Gates 1:18:59

I'll remember some of it. When

Charles Levine 1:19:01

I say all of it, I mean, are you gonna? Does your brain just certain concepts? When I build them, I don't forget them, but when someone else builds them, it takes a little bit. But since you built this, you should remember this. Well, right?

Alex Gates 1:19:14

I probably

Charles Levine 1:19:14

will, because the reason I'm saying is I don't want to let

Alex Gates 1:19:18

this go, right?

Charles Levine 1:19:19

Yeah. Well, okay, great,

Alex Gates 1:19:20

you have the transfer.

Charles Levine 1:19:21

This is, again, one of the best every teacher.